Llama 3.2

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> **Demo Environment**

Development Board: Jetson AGX Orin series motherboard

Meta Llama 3.2 is a series of advanced open-source large-scale language models (LLMs) developed by the Meta AI department.

1. Model Size

Model	Size
llama3.2:1b	1.3GB
llama3.2:3b	2.0GB

2. Performance

Lightweight instruction-tuned benchmarks

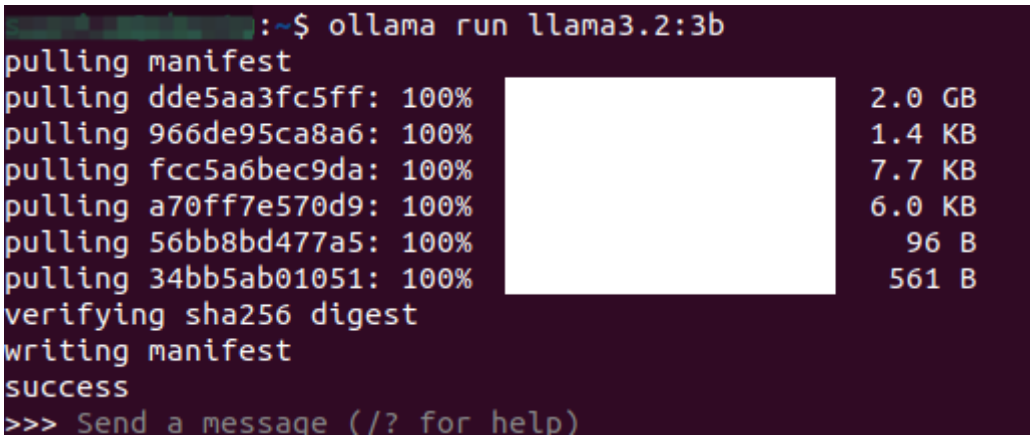
Category Benchmark	Llama 3.2 1B	Llama 3.2 3B	Gemma 2 2B IT (measured)	Phi-3.5-mini IT (measured)
General MMLU (5-shot)	49.3	63.4	57.8	**69.0**
Open-rewrite eval (0-shot, rougeL)	**41.6**	40.1	31.2	34.5
TLDR9+ (test, 1-shot, rougeL)	16.8	**19.0**	13.9	12.8
IFEval	59.5	**77.4**	61.9	59.2
Tool Use BFCL V2	25.7	**67.0**	27.4	58.4
Nexus	13.5	**34.3**	21.0	26.1
Math GSM8K (8-shot, CoT)	44.4	77.7	62.5	**86.2**
MATH (0-shot, CoT)	30.6	**48.0**	23.8	44.2
Reasoning ARC Challenge (0-shot)	59.4	78.6	76.7	**87.4**
GPQA (0-shot)	27.2	**32.8**	27.5	31.9
Hellaswag (0-shot)	41.2	69.8	61.1	**81.4**
Long Context InfiniteBench/En.MC (128k)	38.0	63.3	—	39.2
InfiniteBench/En.QA (128k)	20.3	19.8	—	11.3
NIH/Multi-needle	75.0	84.7	—	52.7
Multilingual MGSM (0-shot, CoT)	24.5	**58.2**	40.2	49.8

3. Using Llama 3.2

3.1 Running Llama 3.2

Use the run command to run the model. If the model is not already downloaded, it will automatically pull the model from the Olama model library:

```
ollama run llama3.2:3b
```



3.2 Starting a Conversation

```
How many minutes are there in a day?
```

Response time depends on your hardware configuration, so please be patient!

```
>>> How many minutes are there in a day?
There are 1440 minutes in a day. This is calculated by multiplying the
number of hours in a day (24) by the number of minutes in an hour (60).

24 hours/day * 60 minutes/hour = 1440 minutes/day

>>> Send a message (/? for help)
```

3.3 Ending a Conversation

Use the `Ctrl+d` shortcut or `/bye` to end a conversation!

References

Ollama

Official Website: https://ollama.com/

GitHub: https://github.com/ollama/ollama

Llama 3.2

Official Website: https://www.llama.com/docs/model-cards-and-prompt-formats/llama3_2/

Ollama Model: https://ollama.com/library/llama3.2